

TECHNICAL CALIBRATION ASSIGNMENT

TASK BRIEF: FULLY FUNCTIONAL RAG SYSTEM IMPLEMENTATION — DESIGN AND ANALYSIS OF ALGORITHMS (AKTU)

Target Domain:	AKTU Curriculum — Design & Analysis of Algorithms (DAA)	Assignment Nature:	Technical Engineering Calibration
Deliverable:	Working RAG System + Comprehensive PDF Report	Implementation:	Autonomous Build (Zero Prescriptive Recipes)

MANDATORY CALIBRATION NOTICE: This assignment serves as an engineering calibration task to benchmark technical capability in designing, implementing, and thoroughly verifying an end-to-end Retrieval-Augmented Generation (RAG) system. No architecture blueprints, framework prescriptions, or implementation recipes are provided. You must research, architect, and construct the entire functional system independently.

1. Technical Objective & Scope

You are tasked with engineering a fully functional, autonomous Retrieval-Augmented Generation (RAG) system tailored specifically to the official curriculum and examination requirements of the AKTU subject: **Design and Analysis of Algorithms (DAA)**.

The system must accept complex academic and algorithmic prompts, retrieve precise syllabus context, and generate comprehensive, step-by-step technical explanations, proofs, algorithm traces, and complexity bounds. The entire implementation must be executed on your own without reliance on pre-baked templates.

2. Mandatory Technical Deliverables & System Requirements

2.1 End-to-End Functional RAG Pipeline

You must build a complete working RAG pipeline on your own. You have full architectural freedom regarding data ingestion, document preprocessing, chunking strategy, indexing, embedding generation, vector/text retrieval mechanisms, and LLM orchestration. You must ensure the system can ingest DAA academic material and answer queries reliably.

2.2 Visual Schematics & Diagram Linking by Persistent ID

Algorithmic explanations require visual clarity. Your RAG pipeline must explicitly account for diagrams (e.g., recurrence trees, AVL tree rotations, Red-Black tree rebalancing, Graph search traversals, minimum spanning tree stages, dynamic programming tables, and algorithm flowcharts).

- **Unique ID Mapping:** Every diagram asset must be assigned a persistent, unique identifier (e.g., DIAG_DAA_U3_AVL_ROT_01).
- **LLM Reference Integration:** When the query requires a visual representation, the retriever and LLM must accurately reference and output the exact Diagram ID in the generated answer text so the user interface can retrieve and render the corresponding schematic.
- **Test Coverage:** Verify that answers requiring diagrams correctly emit the corresponding Diagram ID without broken references or mismatches.

2.3 Rigorous Handling of Numericals & Mathematical Proofs

The DAA syllabus is heavily analytical. Your RAG system must be engineered to handle numerical computations and mathematical derivations accurately, including:

- **Recurrence Relations:** Solving recurrences via Master Theorem (all cases), Recursion Tree Method, and Substitution Method with explicit step-by-step substitution and induction steps.
- **Dynamic Programming & Greedy Computations:** Step-by-step matrix/table construction (e.g., 0/1 Knapsack tabular values, LCS length and directional arrows, Floyd-Warshall distance matrices, Matrix Chain Multiplication parenthesization).
- **Complexity Derivations:** Formal proofs for asymptotic bounds (Big-O, Omega, Theta) with concrete constants (c, n^k).

2.4 Cross-Topic Term Disambiguation & Semantic Stress Testing

A critical engineering failure in naive RAG systems is keyword collision across multiple topics. For example, algorithmic concepts or keywords appear in 2, 3, or more disparate units (e.g., *'dynamic programming'* in matrix chain multiplication vs knapsack; *'greedy approach'* in fractional knapsack vs Huffman coding; *'relaxation'* in Dijkstra vs Bellman-Ford; or terms shared across data structures and algorithms).

- **Disambiguation Verification:** When a query contains an ambiguous or overlapping term, test which topic/unit the retriever pulls.
- **Accuracy Audit:** Rigorously evaluate whether the retriever retrieves the intended context and whether the generated answer correctly resolves the semantic context rather than blending unrelated syllabus units.

3. Standardized RAG Testing & Verification Framework

To ensure your RAG system meets professional engineering standards, you must benchmark and validate your system across the industry-recognized **RAG Evaluation Checklist**. Evaluating a RAG system on end-to-end output alone is insufficient; you must evaluate both the **Retrieval Stage** and the **Generation Stage** independently.

Evaluation Stage	Metric / Dimension	Verification Objective & Test Criteria	Pass Criterion
Retrieval Stage	Context Relevance	Measures whether the retrieved chunks are strictly pertinent to the query and free of irrelevant noise.	High relevance; zero unrelated chunks in top-k context.
Retrieval Stage	Context Recall	Verifies that the retrieved context contains all necessary facts, steps, and formulas required to formulate a complete answer.	Context contains 100% of required algorithmic steps.
Retrieval Stage	Context Precision & Ranking	Assesses whether the most critical and directly relevant context chunks are ranked at the top of the retrieved context.	Top-1 / Top-2 chunks contain primary solution substance.
Generation Stage	Faithfulness / Groundedness	Measures whether every claim, step, and formula in the generated response is strictly grounded in the retrieved context.	Zero hallucinations; no unverifiable claims.
Generation Stage	Answer Relevance	Checks whether the generated response directly answers the user prompt without evading technical questions or generating fluff.	Direct, exhaustive answer addressing all question parts.
Generation Stage	Semantic & Mathematical Accuracy	Validates technical correctness of recurrence solutions, step-by-step arithmetic, pseudocode logic, and asymptotic proofs.	100% mathematically sound; exact complexity notation.
Specialized Domain	Cross-Topic Disambiguation Rate	Evaluates retrieval and generation precision when queries contain terms spanning multiple syllabus topics.	System accurately resolves intended topic without collision.
Specialized Domain	Diagram ID Linkage Rate	Verifies that visual-dependent answers emit the exact Diagram ID to enable downstream image rendering.	Correct Diagram ID emitted for all visual queries.
Specialized Domain	Negative / Out-of-Scope Robustness	Tests the system's ability to gracefully refuse or acknowledge lack of information when given out-of-scope or unanswerable queries.	Explicit rejection without hallucinated answers.

4. Submission Deliverable: Formal PDF Evaluation Report

Upon completing the build and executing the test suite, you must compile and submit a comprehensive **PDF Evaluation Report**. Your report must explicitly check all boxes of the evaluation checklist and contain the following structured sections:

- System Architecture Overview:** Concise summary of the RAG pipeline you designed and implemented (indexing strategy, chunking boundaries, retrieval model, and LLM prompt design).
- Diagram & Asset ID Index:** Catalog of diagram assets indexed, showing how Diagram IDs are formatted and mapped to algorithm concepts.
- Numerical Handling Demonstration:** Explicit walk-through of at least 3 numerical test cases (e.g. Master Theorem derivation, Knapsack table filling, or Dijkstra tracing) showing exact generated steps.
- Cross-Topic Disambiguation Stress Test:** Detailed analysis of multi-topic keyword collisions (e.g. testing queries with ambiguous terms), showing which context was fetched and verifying answer accuracy.
- End-to-End Test Case Logs:** At least 10 authentic test query-response pairs including: Query Prompt, Retrieved Context Chunks (with scores if applicable), and Raw Generated Output.
- Quantitative Evaluation Scorecard:** A structured scorecard grading your system against the 9 checklist metrics in Section 3 (Context Relevance, Context Recall, Context Precision, Faithfulness, Answer Relevance, Semantic Accuracy, Disambiguation Rate, Diagram ID Rate, Robustness).
- Failure Mode Analysis & Limitations:** Objective discussion of edge cases where the system failed, why the failure occurred (retrieval vs generation), and how it can be mitigated.

DELIVERABLE SPECIFICATION: Compile your evaluation results into a single formal PDF document. Ensure all test prompts, raw generated responses, diagram IDs, and numerical calculations are clearly legible. Submit the final PDF report along with instructions to run your working RAG pipeline.

Assigned Task ID:	CALIB-RAG-AKTU-DAA-01	Target Subject:	Design and Analysis of Algorithms
Evaluation Mode:	Empirical Metric-Gated Audit	Expected Report Format:	Standard PDF Document